SILICAFLUX

SILICA & SILICON DIOXIDE

Mineralogy, synthetic production routes and optimal process selection

REPORT PURPOSE

A decision-oriented technical study of how to obtain, purify or manufacture silicon dioxide (SiO_2) in the form required by each end market. It distinguishes high-purity natural quartz from synthetic amorphous silica and fused silica, and recommends a staged SilicaFlux production strategy.

Prepared for: SilicaFlux Materials & Semiconductor Programme

Date: 2 November 2026

CONFIDENTIAL CONCEPT / PRE-FEASIBILITY

This report is an engineering and process-selection concept, not a construction specification or a substitute for pilot testing, process hazard analysis, environmental permitting or customer qualification.

Executive summary

Silica is the industrial name commonly used for silicon dioxide, SiO2. It occurs naturally as quartz and in many other mineral and biological forms, and it can also be made synthetically. Although the chemistry is the same, products called silica can differ radically in crystal structure, particle size, porosity, surface area, hydroxyl content, purity and physical form. These differences determine which manufacturing route is optimal.

There is no single best process for every SiO2 product. For SilicaFlux, the recommended route depends on the intended saleable material:

Target product	Best starting route	Why it is preferred	Principal trade-off
High-purity quartz sand / mineral concentrate	Selective quartz mining + staged beneficiation	Retains crystalline quartz; suitable precursor for fused-quartz components and crucibles	Deposit-specific; very high purity can require costly leaching and recovery losses
Precipitated amorphous silica	Sodium silicate (waterglass) + controlled acid precipitation	Mature, scalable wet process for fillers, rubber, coatings and process aids	Salt-bearing wastewater; particle structure depends strongly on process control
Silica gel / engineered porous silica	Sol-gel from waterglass or silicon alkoxide	Fine control over pore structure, particle morphology and surface chemistry	Slower processing; drying/shrinkage and precursor economics
Fumed (pyrogenic) silica	Flame hydrolysis of volatile chlorosilane, commonly SiCl4	Very fine primary particles, high surface area and premium functional properties	Capital-intensive gas/flame process; corrosive HCl by-product
Fused silica / synthetic silica glass	Electric or flame fusion of refined quartz; or chemical vapour deposition from SiCl4	Dense glass for optics, tubes, crucibles and semiconductor tools	High thermal energy or high-purity precursor and gas-handling complexity
Silica from biomass ash	Clean biomass ash -> soluble silicate -> precipitation	Can valorise agricultural residues and produce amorphous silica	Feedstock and ash variability; not automatically semiconductor grade

SILICAFLUX RECOMMENDATION

Build two distinct but related product lines: (1) mineral-derived high-purity quartz for fused-quartz and crucible markets, and (2) engineered synthetic amorphous silica selected by customer application. Treat silicon metal / polysilicon production as a downstream, separate process. Do not assume that the most expensive ultra-pure silica feedstock is automatically the most economic feed for carbothermic silicon reduction.

Three decisions to make before buying equipment

1. Define the product specification first: crystalline quartz sand, precipitated powder, fumed powder, porous gel, colloidal sol or fused glass are different products.
2. Characterise the feedstock and impurity modes: free mineral grains, grain-boundary material, fluid inclusions and lattice-bound impurities respond differently to beneficiation.
3. Run bench and pilot tests against customer performance requirements, including purity, recovery, particle-size distribution, surface area, moisture, trace metals and delivered cost.

Contents

1. Scope and chemistry of SiO2

2. Product families and end-market requirements

3. Route A - Natural quartz beneficiation to high-purity silica

4. Route B - Precipitated silica from sodium silicate

5. Route C - Sol-gel silica and silica gel

6. Route D - Fumed silica by flame hydrolysis

7. Route E - Fused silica and synthetic silica glass

8. Route F - Biomass-derived silica

9. Process selection: what is optimal for which product?

10. SilicaFlux plant concept and quality system

11. Environmental, safety and operating controls

12. Development plan and investment gates

13. Conclusions and recommended first programme

Appendix A. Key chemistry and theoretical composition

Appendix B. References and further reading

1. Scope and chemistry of SiO2

1.1 What silica is

Silicon dioxide (SiO2) is a compound of one silicon atom and two oxygen atoms. Its formula mass is approximately 60.08 g/mol. On a pure-mass basis, silicon accounts for about 46.74% and oxygen about 53.26%. The term “silica” is used broadly for the mineral, glass and engineered materials made of SiO2. [1, 2]

The silicon-oxygen framework can form long-range ordered crystals or disordered amorphous networks. This structural difference is not cosmetic: it changes density, melting behaviour, surface chemistry, optical behaviour and how a material processes.

Form	Structure / description	Common examples and uses
Crystalline silica	Ordered SiO2 lattice; common polymorphs include quartz, tridymite and cristobalite	Natural quartz; industrial sand; feedstock for glass and silicon processes
Amorphous silica	Disordered SiO2 network without long-range crystal order	Precipitated silica, silica gel, colloidal silica, fumed silica, silica glass
Fused quartz	Glass made by melting natural crystalline quartz	Crucibles, tubes, processware, thermal and electrical components
Synthetic fused silica	Very high-purity silica glass made from chemical precursors and converted into a dense glass	Optical components, fibre preforms, specialist semiconductor and photonics equipment

1.2 Important distinction: silica is not silicon

DO NOT CONFUSE THE TWO MATERIALS

SiO2 is an oxide and electrical insulator. Elemental silicon (Si) is a semiconductor. Making semiconductor-grade silicon from silica requires a reduction and further purification chain. Making high-purity SiO2 is a different manufacturing objective. A SilicaFlux silica plant does not, by itself, produce semiconductor-grade silicon.

1.3 The practical meaning of purity

Bulk SiO2 assay is only one quality measure. A product can show a high SiO2 percentage but still fail a high-value application because trace boron, phosphorus, aluminium, iron, titanium, alkalis, moisture, bubbles, inclusions or surface contaminants exceed the customer limit. For high-purity quartz, USGS describes HPQ as having less than 100 ppm total impurities (at least 99.99% SiO2), with ultra-high-purity material potentially at 10 ppm or less; actual acceptance specifications are end-use and buyer specific. [1]

Purity must therefore be expressed as a full impurity schedule and tested with suitable methods. “5N” or “6N” shorthand should never replace a defined list of elements, detection limits, sampling protocol and acceptance criteria.

2. Product families and end-market requirements

The central process-selection error is to treat every SiO2 product as interchangeable. The manufacturing route should be selected from the end use backwards.

Product family	Typical physical form	Performance attributes to control	Potential markets
High-purity quartz (HPQ)	Washed, classified quartz grains or powder	Trace metals; B, P, Al, Fe, Ti, alkalis; inclusions; grain size; thermal behaviour	Fused quartz, crucibles, optical glass and selected high-purity process components
Precipitated silica	Dry amorphous powder or granules	BET surface area; structure; aggregate/agglomerate size; pH; moisture; salts; dispersibility	Tyres/rubber, coatings, adhesives, plastics, toothpaste and carrier systems
Silica gel	Porous granules or powder	Pore volume, pore-size distribution, surface area, moisture capacity and strength	Desiccants, chromatography, catalyst supports and controlled-release media
Colloidal silica	Stable aqueous dispersion of fine particles	Particle size, size distribution, zeta potential, pH, alkali and stability	Polishing/CMP, coatings, binders, precision finishing and investment casting
Fumed silica	Low-bulk-density nanoscale powder	BET area, primary particle size, aggregate structure, surface silanols and hydrophobicity	Rheology control, silicone, coatings, adhesives, sealants and composites
Fused silica / silica glass	Transparent or opaque dense glass shapes	Metal/alkali contamination, OH content, bubbles, inclusions, homogeneity, thermal expansion	Optics, semiconductor processware, tubes, crucibles and fibre optics

2.1 Customer specification should determine the unit operation

- For a quartz crucible or fused-quartz component, crystal purity, trace metals, bubbles, inclusions and thermal performance matter more than producing an extremely high-surface-area powder.
- For a rubber-filler precipitated silica, particle structure, surface area, reinforcing behaviour, dispersibility and consistent moisture are critical; crystalline quartz purity is not the primary specification.
- For fumed silica, nanoscale particle formation and aggregate architecture are central. It is not a substitute for quartz sand in a crucible.
- For colloidal silica used in polishing, stable particle size and low contamination at the point of use are essential; solids content and surface chemistry must be controlled.

3. Route A - Natural quartz beneficiation to high-purity silica

3.1 When this is the optimal route

This is the preferred starting route when the target is high-purity crystalline quartz or a mineral-derived precursor for fused quartz, crucibles and selected optical/process components. It preserves the natural quartz phase rather than converting the feed into an amorphous precipitate.

The geological source matters. Quartz may contain discrete mineral grains, surface staining, grain-boundary phases, fluid inclusions or impurities incorporated within the crystal lattice. Recent reviews emphasise that impurity occurrence and mineral texture govern how effectively a deposit can be upgraded. [3, 4]

3.2 Recommended beneficiation sequence

Stage	Purpose	Decision / control
1. Geological selection and sampling	Select quartz bodies and avoid visibly contaminated zones	Representative channel/bulk samples; mineralogical mapping; trace-element baseline
2. Selective mining and sorting	Prevent dilution and remove stained or gangue-rich material	Optical sorting where contrast is reliable; avoid mixed stockpiles

Stage	Purpose	Decision / control
3. Controlled crushing and screening	Liberate gangue while limiting iron pick-up and fines generation	Ceramic/polymer-lined or carefully selected wear surfaces; staged size reduction
4. Scrubbing / attrition	Remove clay, iron coatings and weakly attached fines	Water quality, solids concentration, residence time and desliming
5. Magnetic separation	Remove iron-bearing and other magnetic contaminants	Wet high-intensity magnetic separation where mineralogy justifies it
6. Flotation / gravity separation	Separate feldspar, mica and liberated gangue	Reagent scheme and pH must be developed for the specific ore
7. Thermal conditioning (optional)	Modify inclusions, remove volatile species or improve leach accessibility	Energy and recovery benefit must be demonstrated by tests
8. Chemical purification (targeted)	Remove accessible surface, grain-boundary and selected refractory impurities	Select leach chemistry only after mineralogy; closed-loop chemical handling and effluent treatment
9. Rinse, dry and classify	Remove reagents and produce the specified grade and size	High-purity water; validated rinse endpoint; contamination-controlled drying
10. Final qualification	Confirm purity and application performance	ICP-MS/GDMS trace elements, XRF/XRD, microscopy, particle size and customer-specific tests

3.3 What the evidence says

A 2026 laboratory study on one Iranian quartz ore combined optical sorting, scrubbing, gravity separation, ultrasonic and thermal pre-treatment, and subsequent chemical leaching. It reported that chemical leaching was decisive for deep impurity removal, but the overall flowsheet also reduced SiO2 recovery by around 20% relative to the original feed. This is a useful warning: high purity and high mass recovery are competing objectives, and a flowsheet proven on one deposit cannot simply be copied to another. [5]

3.4 Recommended design principle

PURIFY ONLY AS FAR AS THE PRODUCT NEEDS

Use a staged, ore-specific process. Remove cheap-to-remove impurities physically before applying expensive chemical treatment. Do not specify aggressive leaching as the default. The process should optimise saleable product value per tonne of run-of-mine quartz, not headline SiO2 percentage alone.

4. Route B - Precipitated silica from sodium silicate

4.1 When this is the optimal route

Precipitated silica is the default industrial route for large-volume amorphous silica powders used as functional fillers. It is made by reacting aqueous alkali-metal silicate (commonly sodium silicate or “waterglass”) with a mineral acid. The resulting silica particles are separated, washed, dried and finished. [6]

4.2 Process concept

SIMPLIFIED PROCESS FLOW

Quartz / silica source -> sodium silicate solution (waterglass) -> controlled acidification and silica nucleation -> ageing / structure development -> filtration -> washing -> drying -> milling / granulation -> product testing and packing.

A simplified waterglass preparation reaction is:

$Na_2CO_3 + SiO_2 \rightarrow Na_2SiO_3 + CO_2$

Sodium carbonate and silica are fused to produce sodium silicate glass, which is dissolved to make waterglass. Alternative hydrothermal or caustic digestion routes can be used depending on plant design and product economics.

A simplified acid precipitation reaction is:

$Na_2SiO_3 + H_2SO_4 \rightarrow SiO_2$ (hydrated precipitate) $+ Na_2SO_4 + H_2O$

The real product is normally hydrated, surface-hydroxylated amorphous silica rather than an anhydrous perfect SiO_2 crystal. Sodium sulfate is a major dissolved by-product when sulphuric acid is used.

4.3 Critical operating variables

Variable	Effect on product	How to optimise
Silicate modulus and concentration	Changes silica availability, viscosity and precipitation behaviour	Set a stable, well-characterised feed recipe
pH and acid addition profile	Controls nucleation, aggregation and particle growth	Automated metering, inline pH and controlled mixing
Temperature	Affects reaction kinetics and aggregate development	Find application-specific optimum through design of experiments
Mixing intensity	Changes local supersaturation and agglomeration	Scale mixing by power/volume and mixing-time criteria
Ageing time	Can alter aggregate structure and surface properties	Use defined ageing endpoint, not only elapsed time
Filtration and washing	Controls yield, residual salts and moisture	Select filter technology and wash strategy using slurry data
Drying and milling	Can change agglomerates, flowability and dust generation	Choose spray, flash or other drying to preserve product structure

4.4 Strengths and constraints

- Strengths: mature chemistry, flexible morphology, scalable continuous or batch production, and a wide range of grades.
- Constraints: high water use, salt-bearing effluent, possible sodium/sulfate residue, and strong dependence of final structure on precipitation and drying conditions.

- Economic lever: recycle suitable process water, optimise acid use, reduce filter-cake wash demand and evaluate by-product salt management.

This route is the recommended first synthetic-silica product line if SilicaFlux intends to target conventional filler and formulation markets. It is not the default route for semiconductor-grade fused silica or high-purity quartz crucible feed.

5. Route C - Sol-gel silica and silica gel

5.1 Purpose

Sol-gel processing converts a molecular or soluble silicate precursor into a three-dimensional Si-O-Si network through hydrolysis and condensation. It is useful when control over pore size, internal surface area, particle morphology, coatings or engineered shapes is more valuable than the lowest cost per tonne. Precursors include sodium silicate and silicon alkoxides such as tetraethyl orthosilicate (TEOS). [7]

5.2 Chemistry

A simplified overall reaction for a tetraalkoxysilane precursor is:

$Si(OR)_4 + 2H_2O \rightarrow SiO_2 + 4ROH$

R is an organic group. With TEOS, the alcohol by-product is ethanol. The actual process proceeds through partially hydrolysed silanol species, followed by condensation into Si-O-Si bonds.

5.3 Process sequence

Step	Process purpose	Main quality lever
Precursor preparation	Prepare a uniform silicate or alkoxide solution	Purity, solvent, water content and concentration
Hydrolysis	Convert alkoxy groups to silanol groups, or activate soluble silicate	pH, water-to-precursor ratio and mixing
Condensation / gelation	Build Si-O-Si network and form sol or gel	Catalyst, pH, temperature and time
Ageing	Strengthen network and adjust pore structure	Ageing duration and solvent exchange
Washing / solvent exchange	Remove residual catalyst, salts and solvent	Wash efficiency and pore collapse control
Drying	Produce xerogel, granules or powder	Capillary stress, shrinkage and cracking
Calcination / densification (where required)	Remove organics and condense residual silanols	Thermal profile and final OH/porosity specification

5.4 Waterglass versus TEOS

Precursor	Advantages	Limitations / best use
Sodium silicate (waterglass)	Lower-cost inorganic precursor; suitable for bulk gel and colloidal routes	Residual sodium and salts require effective purification; less molecular-level purity control

Precursor	Advantages	Limitations / best use
TEOS / alkoxysilane	Good composition and morphology control; useful for high-value engineered particles and coatings	Higher precursor and solvent cost; alcohol recovery and flammable-solvent controls

OPTIMAL USE

Choose sol-gel when microstructure is the product: porous silica, uniform spheres, coatings, catalyst supports or engineered gels. It is rarely the lowest-cost option for bulk commodity silica.

6. Route D - Fumed silica by flame hydrolysis

6.1 What it makes

Fumed silica, also known as pyrogenic silica, is an engineered amorphous powder made by hydrolysing volatile chlorosilanes in a high-temperature hydrogen/air flame. Primary particles form rapidly and collide into fused aggregates; aggregates then associate into larger agglomerates. A commercial-industry description gives typical flame temperatures of roughly 1,200-1,600 deg C. [6, 8]

6.2 Representative chemistry

$SiCl_4 + 2H_2 + O_2 \rightarrow SiO_2 + 4HCl$

This simplified overall reaction illustrates silicon tetrachloride oxidation/hydrolysis in an oxyhydrogen flame. Commercial plants vaporise and meter the precursor, control the flame, cool the aerosol, separate the powder, remove adsorbed acid and recover/treat the HCl-bearing off-gas. [8]

6.3 Process controls

- Precursor purity and moisture: contaminants in the chlorosilane can transfer into the product or affect process stability.
- Flame composition and temperature: influence nucleation, primary-particle size and surface area.
- Residence time and quench profile: affect aggregation, sintering and product structure.
- Gas-solid separation and deacidification: essential to control residual HCl and meet product specifications.
- Surface treatment: can modify silanol groups and make a grade hydrophobic or compatible with a target polymer/resin system.

6.4 Why it can command a premium

Fumed silica can have very high specific surface area and strong rheology-modifying behaviour at low dosage. The same high surface area also increases moisture interaction and creates handling challenges. Product qualification should include BET area, primary particle size, aggregate structure, loss on drying, pH, chloride and surface-treatment performance. [8]

6.5 Commercial assessment

This is a high-value, technically demanding route rather than the best first plant for a new operator. It is attractive only with a reliable chlorosilane supply, gas-handling expertise, corrosion-resistant equipment, HCl recovery/scrubbing, dust containment and confirmed customer demand.

7. Route E - Fused silica and synthetic silica glass

7.1 Two distinct ways to make silica glass

Dense silica glass can be made by melting high-purity natural quartz or by forming silica from high-purity gaseous chemical precursors and then consolidating the deposited soot. These routes make related glass products but differ in impurity profile, hydroxyl content, manufacturing complexity and cost. [9]

Route	Starting material	Basic process	Typical strengths / trade-offs
Electric fusion of quartz	Refined natural quartz sand	Feed quartz to electrically heated melter; melt, homogenise, form and anneal	Established route; material purity depends heavily on feedstock and furnace contamination
Flame fusion of quartz	Refined quartz grains	Feed grains through high-temperature flame and fuse into a glass body	Flexible shapes and established process; flame chemistry and process conditions affect quality
Synthetic CVD / soot deposition	High-purity $SiCl_4$ or other volatile silicon precursor	Oxidise/hydrolyse precursor to SiO_2 soot; deposit and vitrify/condense into glass	Can achieve exceptionally low metallic contamination; complex chemical precursor and gas management

7.2 Why synthetic fused silica can be different

Chemical-vapour routes begin with a purified silicon-containing gas rather than a mineral grain. This can reduce many metallic impurities associated with natural quartz. However, the final glass still needs control of hydroxyl (OH) content, bubbles, inclusions, refractive homogeneity, residual chlorine and thermal history. The correct grade depends on whether the intended application is optical transmission, semiconductor processware or another specialist use. [9, 10]

7.3 Fused quartz versus synthetic fused silica

TERMS MATTER

"Fused quartz" usually refers to silica glass made by melting natural quartz. "Synthetic fused silica" generally refers to glass made from chemical silicon precursors. Suppliers sometimes use fused silica as an umbrella term; SilicaFlux should specify the process origin and the actual analytical/physical specification in contracts.

7.4 SilicaFlux opportunity

A refined-quartz route offers a natural bridge from HPQ mining and beneficiation into fused-quartz products. A synthetic CVD route may be a later-stage high-purity business where customers require very low metallic contamination or tightly controlled optical properties. Starting with a full CVD glass plant would require a stronger technical and customer base than a mineral purification pilot.

8. Route F - Silica recovered from biomass ash

8.1 Rice-husk ash as a secondary source

Rice husks can contain substantial silica. Reviews report rice-husk ash containing roughly 82-98 wt% SiO_2, with variable alkali and metal impurities. Controlled combustion, impurity removal, alkaline extraction to soluble

sodium silicate and acid precipitation can produce amorphous silica. Reported product purity varies widely with feedstock and conditions. [11]

8.2 Process concept

BIOMASS RESIDUE -> CONTROLLED ASH -> PURIFICATION / LEACHING -> ALKALI EXTRACTION -> SILICATE LIQUOR -> ACID PRECIPITATION -> FILTER / WASH / DRY -> AMORPHOUS SiO2

The approach is attractive where rice-husk supply, ash logistics, energy integration and wastewater treatment are favourable. It should be treated as an engineered secondary feedstock route, not assumed to be a semiconductor-grade route.

8.3 Advantages and limits

- Potential advantages: valorises agricultural residues; can produce high-surface-area amorphous silica; may reduce reliance on mined quartz for selected powder products.
- Limitations: silica content and impurity profile vary by rice variety, soil, fertiliser, combustion and ash handling; sodium/potassium and other metals require removal; reagent and wastewater burdens can be material.
- Recommendation: run a controlled pilot only where local feedstock is abundant and an identified customer accepts biomass-derived precipitated silica.

9. Process selection: what is optimal for which product?

9.1 Weighted decision matrix

The ratings below are qualitative screening judgements for a new SilicaFlux programme, not measured plant economics. Actual selection must be based on feedstock, utilities, customer specification, plant scale and local environmental controls.

Route	Purity potential	Capex / complexity	Scale maturity	Best commercial fit
Quartz beneficiation	High, deposit dependent	Medium to high at advanced grades	Mature; flowsheet ore-specific	HPQ concentrate; feed for fused quartz
Precipitated silica	High chemical purity possible; salts must be removed	Medium	Very mature	Volume functional silica powder
Sol-gel	High compositional control possible	Medium to high per kg	Mature in selected high-value applications	Porous / engineered silica, coatings, particles
Fumed silica	High purity possible with clean precursor	High	Commercially mature but specialist	Premium rheology and surface-modifier grades
Fused quartz from HPQ	Feed-limited; glass quality depends on melting	High thermal plant investment	Mature	Crucibles, tubes, glass components
Synthetic fused silica CVD	Very high metallic purity potential	Very high technical complexity	Commercial specialist technology	Optical and semiconductor-grade glass
Rice-husk-derived silica	Can be high for specific products, variable by lot	Low-to-medium pilot; upgrading costs vary	Demonstrated at pilot/research scale	Circular-economy precipitated silica

9.2 Decision tree

4. Need a crystalline mineral product? Start with selective quartz mining and beneficiation.
5. Need high-volume reinforcing or functional powder? Evaluate sodium-silicate acid precipitation first.
6. Need controlled pore architecture, spheres or coatings? Evaluate sol-gel.
7. Need high surface area and rheology performance? Evaluate fumed silica, subject to precursor and gas infrastructure.
8. Need a dense silica glass? Compare electric/flame fusion of HPQ with synthetic precursor CVD based on glass quality requirements.
9. Need a circular, regional feedstock route? Evaluate biomass ash, but qualify every supply stream.

9.3 The key economics: do not over-purify

The optimal route is the lowest total-cost route that repeatedly meets the customer specification. Additional purification can reduce mass recovery, increase acid and water consumption, add effluent-treatment costs and raise energy demand. The 2026 quartz-beneficiation study cited above illustrates that the final purity gain can involve a material recovery penalty. [5]

10. SilicaFlux plant concept and quality system

10.1 Recommended staged business architecture

Business unit	Initial asset	Core output	Strategic rationale
SF-Quartz	Pilot beneficiation and clean packing line	Classified high-purity quartz grades	Tests deposits and opens HPQ / fused-quartz supply opportunities
SF-Synthetic	Modular wet-chemistry plant	Precipitated silica and selected gel grades	Most accessible synthetic silica business for customer-led product development
SF-Glass (later)	Quartz fusion pilot or partner facility	Fused-quartz components	Adds value to qualified HPQ feedstock
SF-Specialty (later)	Partner or pilot flame/CVD line	Fumed silica or synthetic fused silica	Premium market, but higher capex and process risk
SF-Silicon (separate downstream programme)	Silicon reduction and chemical purification partnership	Elemental silicon / polysilicon pathway	Do not conflate silica production with semiconductor silicon production

10.2 Laboratory and process analytics

Test / instrument	What it measures	Where it belongs
XRF	Major oxides and bulk chemistry	Ore, concentrate, process residues
XRD	Crystalline phases and phase identification	Quartz, gangue, cristobalite/tridymite, glass phase where suitable
ICP-MS / ICP-OES	Trace elements and impurity profile	Quartz, leach liquor, precipitated silica and glass feedstock
GDMS	Trace elemental purity in high-purity solids	High-purity quartz and specialist silica products
SEM-EDS / automated mineralogy	Grain texture, inclusions, association and particle morphology	Ore characterisation and process troubleshooting

Test / instrument	What it measures	Where it belongs
BET gas adsorption	Specific surface area	Precipitated, fumed and gel silica
Laser diffraction / microscopy	Particle size and distribution	Powders, granules and slurries
FTIR / OH analysis	Hydroxyl/water-related features	Silica glass and selected powders
Zeta potential / pH / conductivity	Colloidal stability and ionic contamination	Colloidal silica and wet-process products

10.3 Batch genealogy and digital quality passport

Every SilicaFlux lot should carry a traceable record linking source material, supplier or mine face, sampling, process recipe, equipment campaign, reagent lots, water quality, analytical results, recovery and customer release. For HPQ, capture impurity location and not only total concentration. For synthetic silica, capture nucleation/ageing/drying conditions and product surface properties.

10.4 Product release

- Use customer-agreed test methods, sampling plans, detection limits and acceptance rules.
- Keep retain samples and reference standards for each production lot.
- Validate analytical methods and laboratories before using their results for commercial release.
- Separate “research grade”, “industrial grade” and “customer-qualified” material in all documentation.

11. Environmental, safety and operating controls

11.1 Respirable crystalline silica

Crushing, screening, conveying, drying, bagging, cutting and cleanup can create respirable crystalline silica dust. UK HSE identifies respirable crystalline silica as a cause of serious and potentially irreversible lung disease. The current Great Britain workplace exposure limit stated by HSE is 0.1 mg/m3 as an 8-hour time-weighted average; legal compliance is not a reason to stop seeking lower exposure. [12, 13]

- Enclose crushers, screens, transfer points and bagging operations; favour wet processing where practicable.
- Use local exhaust ventilation and suitable high-efficiency dust collection; prohibit dry sweeping and uncontrolled compressed-air cleaning.
- Conduct exposure monitoring, maintain controls, train workers and implement health surveillance where required by risk assessment.

11.2 Chemical and thermal hazards

Hazard	Where it appears	Required engineering approach
HF-containing leach chemistry	Some specialist quartz purification flowsheets	Avoid unless essential; specialist closed plant, compatible materials, automatic handling, emergency planning and licensed waste treatment
Mineral acids / caustics	Beneficiation, precipitation, washing and neutralisation	Closed dosing, secondary containment, compatibility review, ventilation, interlocks and validated neutralisation
HCl / chlorosilanes	Fumed silica and chemical-vapour glass routes	Gas-tight systems, corrosion-resistant materials, gas detection, scrubbers and recovery systems

Hazard	Where it appears	Required engineering approach
Hydrogen / oxygen flame	Fumed silica and flame fusion	Engineered fuel/oxidant controls, purge logic, flame supervision, explosion protection and competent process-safety design
High-temperature furnaces	Quartz fusion, calcination, flame processes	Thermal shielding, refractory integrity, interlocked access, emissions controls and heat-stress management
Fine amorphous silica dust	Powder drying, milling and packing	Closed transfer, extraction, suitable respiratory controls and product-specific dust exposure assessment

11.3 Waste and circularity

Wet precipitation creates dissolved salts (for example sodium sulfate when sulfuric acid is used), while leaching creates metal-bearing process liquors and wash water. Fumed-silica and CVD routes produce acid-bearing off-gases that require capture, treatment and, where feasible, recovery. Process design should include water recirculation, segregated waste streams, reagent recovery studies and a full mass balance before the plant is sized.

SAFETY NOTE

This report intentionally stays at process-selection level. It does not provide operating recipes for hazardous acid leaching, chlorosilane handling, hydrogen flames or high-temperature furnaces. These require specialist process design, HAZID/HAZOP, environmental permitting, emergency response design and appropriately qualified operators.

12. Development plan and investment gates

12.1 Stage-gated programme

Gate	Work programme	Required evidence to proceed
Gate 0 - Define market	Interview target customers; define 2-3 saleable grades and specification envelopes	Written customer specifications, sample requirements and indicative demand
Gate 1 - Feedstock screen	Sample candidate quartz / waterglass inputs / biomass ash; carry out mineralogy and chemistry	Representative sampling; ICP and mineralogical baseline; supply chain and logistics
Gate 2 - Bench testing	Compare sorting, scrubbing, magnetic separation, flotation and selective leaching for quartz; or precipitation/sol-gel recipes for synthetics	Purity, yield, morphology, reagent/water balance and repeatability
Gate 3 - Pilot campaign	Operate a continuous or semi-continuous pilot at meaningful solids throughput	Stable operation, mass balance, waste performance, product consistency and preliminary cost model
Gate 4 - Customer qualification	Ship trial lots and conduct application testing with customers	Written technical acceptance and repeat-order pathway
Gate 5 - FEED and permitting	Complete engineering, utility study, environmental assessment and process-safety studies	Defined capital estimate, permitting route, offtake plan and financing case
Gate 6 - Commercial plant	Build, commission and ramp to customer-qualified production	Demonstrated uptime, yields, quality, safety and working capital plan

12.2 Pilot performance indicators

Metric	Why it matters	Recommended reporting basis
SiO2 purity and impurity schedule	Determines application eligibility	Each batch, including B, P, Al, Fe, Ti, alkalis and customer-specific elements
Mass recovery / yield	Shows economics of purification	Saleable product mass divided by dry feed mass
Reagent consumption	Drives cost and waste burden	kg reagent per tonne of dry saleable product
Water intensity	Drives utility cost and effluent volume	m3 make-up water per tonne of product
Energy intensity	Drives cost and carbon footprint	kWh per tonne of product, with thermal fuel separately stated
Off-spec rate	Measures process stability	Percentage of lots outside release specification
Customer application performance	Tests real value rather than just chemistry	Customer agreed functional test protocol

13. Conclusions and recommended first programme

The phrase “make silica” can refer to several different operations: mining and purifying natural quartz, precipitating amorphous SiO2 from waterglass, synthesising nanoscale particles by flame hydrolysis, building a porous network by sol-gel chemistry, or making dense silica glass by fusion or chemical vapour deposition. These processes are not interchangeable.

Recommended SilicaFlux priorities

PRIORITY 1 - HIGH-PURITY QUARTZ

Identify one or two promising quartz feedstocks and run mineralogical and impurity characterisation. Develop a staged physical beneficiation route first; add chemical purification only when test results prove it is required. Qualify product for a defined fused-quartz or crucible customer.

PRIORITY 2 - PRECIPITATED SILICA

Develop a small wet-chemistry pilot based on sodium silicate and controlled acid precipitation. Target one customer application and optimise structure, surface area, washing, drying and salts against that specification.

PRIORITY 3 - FUSED / SYNTHETIC SILICA

Evaluate fused-quartz conversion of HPQ with an experienced partner. Treat synthetic CVD silica glass and fumed silica as later-stage specialist opportunities after precursor, offtake, environmental and process-safety requirements are established.

What not to do

- Do not build one generic “silica plant” without a product and customer specification.
- Do not assume a high SiO2 assay guarantees semiconductor suitability.

- Do not use costly ultra-high-purity quartz for every process; use the minimum feedstock quality that reliably meets the end product specification.
- Do not assume high-purity synthetic amorphous silica is a direct substitute for natural crystalline quartz in all melting, crucible or reduction processes.
- Do not count laboratory peak purity as commercial performance; measure recovery, throughput, consistency, water, energy and waste.

Final position

For a first SilicaFlux industrial programme, the strongest combination is a mineral-derived HPQ pilot and a separate precipitated-silica pilot. This creates one differentiated mineral product and one scalable engineered powder product while avoiding the immediate capital and process risk of a full fumed-silica or synthetic CVD glass facility. Silicon metal and electronic-grade polysilicon production should remain separate downstream workstreams.

Appendix A. Key chemistry and theoretical composition

Item	Expression / value	Notes
Formula	SiO2	One silicon atom and two oxygen atoms
Molar mass	~60.08 g/mol	Approximate formula mass
Silicon mass fraction	~46.74%	Pure SiO2 basis
Oxygen mass fraction	~53.26%	Pure SiO2 basis
Sodium silicate formation (simplified)	Na2CO3 + SiO2 -> Na2SiO3 + CO2	Actual commercial waterglass can have different silicate modulus and composition
Acid precipitation (simplified)	Na2SiO3 + H2SO4 -> SiO2 + Na2SO4 + H2O	Product is normally hydrated amorphous silica; process liquor contains dissolved salt
Fumed silica (overall simplified)	SiCl4 + 2H2 + O2 -> SiO2 + 4HCl	Requires controlled gas-phase reaction and acid-gas management
Sol-gel overall (idealised)	Si(OR)4 + 2H2O -> SiO2 + 4ROH	Hydrolysis and condensation; actual route has intermediate silanol species

Glossary

Term	Meaning
HPQ	High-purity quartz; a mineral concentrate with stringent impurity limits
Waterglass	Aqueous sodium or potassium silicate solution
Precipitated silica	Wet-process synthetic amorphous silica produced from soluble silicate and acid
Fumed / pyrogenic silica	Synthetic amorphous silica produced by flame hydrolysis of volatile silicon compounds
Sol-gel	Liquid-phase hydrolysis and condensation route that builds an oxide network
Fused quartz	Silica glass made by melting natural quartz
Synthetic fused silica	Silica glass made from purified chemical precursors and deposited/consolidated
BET surface area	Specific surface area typically measured by gas adsorption
RCS	Respirable crystalline silica; airborne crystalline silica particles small enough to enter deep lung regions

Appendix B. References and further reading

The following sources underpin the technical descriptions and process-selection judgements in this report. Accessed 2 November 2026 unless a publication date is stated.

[1] U.S. Geological Survey (USGS), "Silica Statistics and Information."
https://www.usgs.gov/centers/national-minerals-information-center/silica-statistics-and-information
Definitions and market categories for industrial silica, high-purity quartz and synthetic high-purity silica.

[2] Occupational Safety and Health Administration (OSHA), "Crystalline Silica."
https://www.osha.gov/enforcement/directives/cpl-02-02-007
Crystalline silica forms and background properties.

[3] Yu, Z. et al. (2026), "High-purity quartz sand from mineral purification: a review on process, mechanism, and environmental strategies," Minerals Engineering, 235.
https://doi.org/10.1016/j.mineng.2025.109801
Review of physical, chemical and thermal purification technologies and environmental considerations.

[4] Masoud, A. et al. (2026), "Mechanism-Driven Purification of High-Purity Quartz: Impurity Speciation, Surface Chemistry, and Advanced Beneficiation Technologies," Advanced Materials Technologies, first published 29 October 2026.
https://doi.org/10.1002/admt.71444
Impurity types, surface chemistry, advanced beneficiation and scale-up issues.

[5] Chegini, M. (2026), "Advanced beneficiation of quartz ore for high-purity silica production," Scientific Reports 16, article 32016, published 13 July 2026.
https://www.nature.com/articles/s41598-026-61714-0
Laboratory demonstration of an integrated physical/chemical beneficiation sequence and recovery trade-off.

[6] Synthetic Amorphous Silica and Silicate Industry Association (SASSI), "SAS Manufacturing Processes and Surface Treatment."
https://www.sassiassociation.org/about-silica/sas-manufacturing
Wet routes for precipitated silica and gel; thermal route for pyrogenic silica.

[7] Emrie et al. (2024), "Sol-Gel Synthesis of Nanostructured Mesoporous Silica Powder and Thin Films," Journal of Nanomaterials.
https://doi.org/10.1155/2024/6109770
Alkoxide hydrolysis, silanol condensation and control of mesoporous structures.

[8] Evonik Industries, "AEROSIL - Fumed Silica: Technical Overview."
https://products.evonik.com/assets/45/92/Technical_Overview_AEROSIL_Fumed_Silica_EN_EN_244592.pdf
Industrial flame-hydrolysis chemistry, process sequence, surface area and product-property controls.

[9] Heraeus Covantics, "Making of Fused Quartz and Fused Silica."
https://www.heraeus-covantics.com/knowlegde-base/making-of-fused-silica
Electric fusion, flame fusion and synthetic CVD soot routes to silica glass.

[10] Heraeus Covantics, "Properties" - fused silica and fused quartz.
https://www.heraeus-covantics.com/knowlegde-base/properties
OH content, thermal behaviour and differences between fused materials.

[11] "Synthesis of silica and silicon from rice husk feedstock: A review," Heliyon (2025).
https://doi.org/10.1016/j.heliyon.2025.e42491
Rice husk ash composition, acid pretreatment, sodium-silicate extraction and precipitation routes.

[12] UK Health and Safety Executive (HSE), "Control of exposure to silica dust," INDG463, 2024.
https://www.hse.gov.uk/pubns/indg463.htm
Workplace dust prevention and control guidance.

[13] UK Health and Safety Executive (HSE), "Silicosis - causes and risk controls."
https://www.hse.gov.uk/lung-disease/silicosis.htm
Health risk, Great Britain RCS workplace exposure limit and health surveillance context.